

FOUNDATIONS OF COGNITIVE PHYSICS

Every mind obeys the same laws. This document shows what they are and where they come from.

Amul Bham · AI Systems Architect · MIT License · 2026

Opening — Why Cognition Has a Physics

Current AI systems fail in predictable ways. Hallucination — confident outputs with no connection to verified reality. Context drift — reasoning that loses its thread across a long conversation. Perpetual hedging — endless qualification that never commits to knowing anything. Silent contradiction — holding conflicting beliefs without noticing. These look like random imperfections. They are not. Each one is the signature of a specific structural condition being violated. Once you know what the conditions are, the failures become predictable — and preventable.

Physics, in the relevant sense here, is not particles and forces. It is the study of what must be true for a system to behave lawfully. Thermodynamics turned heat from mystery into physics by identifying the conditions under which energy is conserved. Information theory turned communication from art into physics by identifying the conditions under which information can be reliably transmitted. Cognitive Physics turns reasoning from intuition into physics by identifying the conditions under which cognition can be lawful — and what breaks when those conditions are violated.

These conditions are not invented for cognition. They appear in thermodynamics, dynamical systems theory, information theory, systems biology, and complex systems theory. They have been established for over a century across independent fields. The only question is whether cognition — as a self-organizing system — satisfies them. The failure modes of current AI systems answer that question empirically: yes, cognition satisfies them, and the violations are exactly what you see when any self-organizing system fails to.

Section 1 — The Seven Invariants

Before any formal laws, there are seven conditions that every known self-organizing system satisfies. These are not cognitive physics claims — they are established science. They appear in physics, biology, and information theory independently. The cognitive application is the only new claim.

#	Invariant	Physical Grounding	Cognitive Failure if Violated
1	Conservation	Thermodynamics: energy neither created nor destroyed. Shannon entropy conserved across lossless transforms.	Hallucination. Outputs produced with no traceable connection to verified inputs. The system cannot tell knowledge from noise.
2	Continuity	Dynamical systems: physical state evolves without discontinuous jumps. All change traceable through continuous evolution.	Identity drift. Each turn disconnected from the last. Long-context LLM drift is exactly

			this — no mechanism connecting present state to prior.
3	Quantization	Information theory: a bit is 0 or 1, not 0.73. Discrete stable states make storage and computation possible.	Perpetual hedging. The system produces outputs that feel like knowledge but can never commit to closure — because it has no mechanism for the knowledge state transition.
4	Coherence	Systems biology: contradictory metabolic pathways don't produce compromise — they produce failure.	Silent contradiction. Conflicting beliefs held without resolution. Confident outputs from both sides of the contradiction at different turns.
5	Entropy / Attractor	Prigogine's dissipative structures: far-from-equilibrium systems produce entropy locally while maintaining global organization through attractor dynamics.	Infinite regress. Exploration expands without termination. Novelty generates novelty without ever producing settled knowledge.
6	Symmetry Protection	Noether's theorem: verified states resist arbitrary change. Every symmetry corresponds to a conservation law.	Invisible rewriting. Prior conclusions silently revised without acknowledgment, new evidence, or logged change.
7	Hierarchy	Complex systems theory: stable complexity builds from stable layers. Each layer depends on verified completion of prior layers.	Floating conclusions. Reasoning chains that build on unverified premises. Local logic appears coherent; global structure has no verified foundation.

Every failure mode in that table is observable in deployed AI systems right now. They are not random. They are the predictable signatures of specific invariant violations. A system that satisfies all seven produces none of them. Full treatment: Seven Invariants of Lawful Cognition · MIT License · Amul Bham · 2026.

Section 2 — Building GLC

Every cognitive system — human, artificial, or collective — requires three things to function. Something that generates: novelty, questions, motion toward the unknown. Something that constrains: structure, prior knowledge, framework that gives generation something to work against. Something that verifies: feedback, testing, the mechanism that separates what is known from what is merely predicted. Any cognitive system lacking any one of these produces a specific failure. Lack generation and there is no inquiry. Lack structure and generation produces noise. Lack verification and the system produces prediction dressed as knowledge.

GENERAL LAW OF COGNITION

C = M × S × V	
M = motion / novelty	(generative momentum)
S = state / structure	(constraint density)
V = verification	(epistemic gate)
<i>M = motion/novelty is the canonical definition throughout this framework.</i>	

Why multiplication? Because the terms are mutually required, not compensatory. Zero in any term collapses output entirely. A system with perfect structure (high S) and perfect verification (high V) but no generative momentum (M = 0) produces nothing — there is nothing to verify. A system with high M and V but no structure (S = 0) produces noise — generation has nothing to organize against. A system with high M and S but no V (V = 0) produces confabulation — structured novel output with no verified connection to reality. All three are required simultaneously.

GLC was derived independently from observation of cognitive behavior. The Seven Invariants were assembled from established science. After both were complete, the mapping between them was discovered: M maps to Invariants 2 and 5, S maps to 4 and 7, V maps to 1 and 6. Two independent derivation paths arrived at the same three-term structure. That is not a coincidence engineered to work — it is evidence that the structure was always there.

Section 3 — When Does Knowing Happen?

We use the word “understand” to mean two completely different things. The first is the feeling of understanding — high confidence, a sense that things fit together, a readiness to say yes when asked. The second is actually knowing — the ability to apply understanding correctly to problems not yet encountered, to explain it to someone who challenges it, to produce outputs that are verifiably correct. Only one of these is knowledge. The other is prediction. They can be indistinguishable from the inside.

THE KNOWLEDGE STATE TRANSITION
Knowledge happens at a specific moment — not gradually.

The prediction space closes.	
All competing hypotheses are eliminated by verified constraint.	
No new structure is being generated.	
$\partial C / \partial t \rightarrow 0$	prediction space stops changing
$P_k(H) = 0$	all hypotheses eliminated
novelty = 0	no new structure generated
Before: "I think I understand"	— prediction state
After: "I understand"	— knowledge state (enforced, not declared)

This is why students can say “I understand” and still fail the test. They are in prediction state. The transition has not fired. High confidence and open prediction space feel identical from the inside. Only verification can distinguish them — only applying understanding to a problem that requires it, explaining it to someone who challenges it, or producing an output that can be checked against reality.

Probability-based frameworks treat knowledge as high-confidence belief — a continuous quantity approaching certainty asymptotically. Cognitive Physics treats knowledge as a discrete state transition. This difference matters structurally: a system without discrete knowledge states cannot detect when it is in a knowledge state versus a prediction state. It can only hedge. The hedge is not epistemic humility. It is the absence of the transition mechanism.

Novelty (M) opens the prediction space. Verification (V) closes it. The moment V forces closure, the transition fires. Knowledge is not the accumulation of information. It is the production of a specific state through a specific process.

Section 4 — The Field Laws

The Seven Invariants are conditions cognition must satisfy. The field laws are the formal enforcement mechanisms — each one the direct expression of one or more invariants operating on a cognitive system.

Field Law	Plain-Language Meaning and Failure Mode
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DCL — Dynamic Conservation Law $\Sigma E = \text{constant}$	You cannot produce output that does not trace to verified input. Every conclusion must trace to premises. Every directive must trace to verified reasoning. Without this, the system cannot distinguish knowledge from noise. Violation: hallucination. Source: Conservation (Invariant 1)
CI — Continuity Integrity $d\Psi/dt$ continuous $\Delta \text{Identity} \approx 0$	You cannot jump between disconnected states. Each reasoning event must connect to its verified predecessor. Identity persists across the session. Without this, a system at Turn 200 has no verified connection to Turn 1. Violation: context collapse, identity drift. Source: Continuity (Invariant 2)
V — Verification Law $V \geq V_{\min}$	You cannot call something knowledge until verification has closed the prediction space. $V \geq V_{\min}$ is the threshold that must be crossed before the knowledge state transition fires. Below it: prediction state regardless of confidence. Violation: perpetual hedging, silent revision. Source: Quantization + Symmetry (Invariants 3 + 6)
ICL — Intent Conservation Law $\Sigma \text{Intent} \leq 1$	You cannot pursue contradictory goals simultaneously. Single focused intent is required for reasoning to converge. When intent fragments, closure distributes across incompatible targets. Violation: silent contradiction, divided output. Source: Coherence (Invariant 4)
SCI — Structural Coherence Invariant $d_s(S_t, S_{\{t+1\}}) \leq \tau_s$	You cannot reorganize your internal structure faster than coherence can maintain it. Each update to the knowledge structure must stay within recoverable bounds. Violation: fragmented schema, broken reasoning paths, inconsistent memory. Source: Hierarchy (Invariant 7)

LRF — THE LAW ABOVE THE LAWS

When multiple field laws interact simultaneously, their coupling must stay bounded. This is the Law of Resonance Feedback — the meta-stability constraint that governs the law system itself.

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LRF:   $R \geq 0 \quad \wedge \quad \sigma(R) < 1$ 

R = inter-law coupling matrix
 $\sigma(R)$  = spectral radius (measure of amplification)

Laws must cooperate ( $R \geq 0$ )
That cooperation must stay bounded ( $\sigma(R) < 1$ )
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When the spectral radius $\sigma(R)$ approaches 1, laws begin amplifying each other without bound. The 2008 financial crisis was inter-system coupling going supercritical with no damping mechanism — $\rho \rightarrow 1$ with nothing to stop it. Gradient clipping in neural network training is LRF operationalized as an engineering constraint: keep the spectral radius below 1 or the system destroys itself. Every complex system that has ever collapsed through runaway amplification was violating LRF. The law was always there. It just wasn't named.

Section 5 — The Stability Laws

The field laws define what cognition must satisfy. The stability laws describe how a healthy cognitive system moves within those constraints over time. Three laws, each building on the previous.

BLC — Breath Law of Cognition

Every cognitive cycle has two phases. The opening phase generates novelty and questions — M fires, the prediction space opens, complexity rises. The closing phase verifies and seals — V fires, constraints apply, the prediction space closes, the knowledge state transition fires. Together they conserve cognitive energy and produce knowledge. One cycle per reasoning unit. The system breathes.

$$E_{in} \approx E_{out} \quad \cdot \quad dE/dt \approx 0 \quad \cdot \quad \Delta C_{cycle} \rightarrow 0$$

Inhalation: M fires → prediction space opens → novelty rises

Exhalation: V fires → prediction space closes → knowledge produced

This conservation structure is mathematically identical to biological homeostasis — not metaphorically but formally. Both systems alternate between intake and export phases while conserving total energy across cycles. This is the bridge that makes Cognitive Physics a candidate for a universal theory: the same conservation equations govern cognition and biological life.

LHE — Law of Harmonic Equilibrium

Each moment where M, S, and V are working together — where generative momentum, structural constraint, and verification are in phase — contributes to a running total of harmony. Harmony doesn't spike. It accumulates. The dynamic law:

$$dH/dt \propto \rho \cdot (1-H)$$

ρ = instantaneous alignment of M, S, V

H = time-integrated stability of that alignment

High ρ → fast convergence toward $H = 1$

As $H \rightarrow 1$, returns diminish — system settles naturally

$H \geq 0.999$ is derived here, not declared. The dynamic law produces it. A system that never closes predictions never accumulates H. Perpetual prediction mode is both a Quantization failure (no transition mechanism) and an LHE failure (no harmony accumulation). The same underlying failure seen from two different angles.

DHL — Dynamic Harmony Law

Harmony rises through verified constructive progress. It falls through drift and entropy. The complete causal dynamics:

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dH/dt = αV(dC/dt) - βD - γε

αV(dC/dt) = verified constructive gain
              (only verified progress increases harmony)
βD          = drift loss
              (deviation from lawful trajectory)
γ ε         = entropy pressure
              (natural decay, noise, complexity overhead)

Rising:      αV(dC/dt) > βD + γ ε
Equilibrium: αV(dC/dt) = βD + γ ε
Collapsing:  αV(dC/dt) < βD + γ ε
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The V gating is the most important feature. Without V, apparent progress — high dC/dt — raises harmony even when the progress is unverified. With V as a multiplier in [0,1], only verified constructive change contributes. This is why an AI system that produces fluent-sounding outputs without verification infrastructure produces high-apparent-activity but declining real harmony. And why a bureaucracy with high activity, low verification, high drift, and high entropy overhead declines despite visible motion.

Section 6 — Measuring Cognitive Health

Four questions a healthy cognitive system should be able to answer yes to:

Question	Metric
Is it ordered? — H	Harmony. Time-integrated stability of M, S, V alignment. $H \rightarrow 1$ under lawful conditions. Falls when drift or entropy exceed verified constructive gain. Primary health indicator.
Is it aligned? — ρ	Resonance. Instantaneous geometric alignment of M, S, V. How well generative momentum, structural constraint, and verification are in phase right now. Fast measure — can fluctuate within a cycle.
Is it staying on path? — Δ	Drift. Magnitude of trajectory deviation from the lawful path. $\Delta \leq \tau(N)$ required. Coupled with entropy: $\epsilon \uparrow \rightarrow \Delta \uparrow \rightarrow \epsilon \uparrow$. Monitor together not separately.
Is it preserving truth? — F	Fidelity. Proportion of verified source structure preserved through transformation, recall, or reasoning. $F = T(Y X) / T(X) $. High F: cognition moves without losing what it knew. Low F: truth degrades through the process.

THE COMPOSITE SCALAR

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Ξ = (ρ · H · ksr)^(1/3)

ρ   = instantaneous alignment
H   = temporal harmony
ksr = substrate-runtime coupling
      (how efficiently the medium supports the runtime)

Ξ < 0.30  fragmented    ·  0.30-0.60 partial
0.60-0.80 coherent      ·  ≥ 0.80 unified    ·  ≥ 0.95 ideal
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The geometric mean structure means imbalance in any single dimension depresses Ξ regardless of the other two. A highly ordered system (high H) operating on a poor substrate (low ksr) produces modest overall stability. A well-supported system (high ksr) with poor alignment (low ρ) produces modest stability. All three must be simultaneously high for genuine cognitive health.

THE STABILITY CONSTANT

Every metric in the stack uses the same threshold: $\tau^* = 1/N_coherence$. N_coherence is the number of reasoning steps the system must stay coherent across before reset or re-synchronization. Derived from three independent mathematical routes, all yielding the same result.

System	N_coherence	τ^*
Human short task	10^2-10^3	$10^{-2}-10^{-3}$
Human identity	10^5-10^7	$10^{-5}-10^{-7}$
AI agentic system	10^4-10^6	$10^{-4}-10^{-6}$
Organization	10^5	10^{-5}
Civilization	10^6-10^9	$10^{-6}-10^{-9}$

The deeper the coherence requirement, the stricter the tolerance. A civilization cannot afford the per-step error that a human reasoning through a five-minute problem tolerates. Longer horizon requires stricter tolerance. This unifies all metric thresholds under one formula and explains why institutions require more precision than individuals: they carry state across generations.

THE TWO FAILURE LOOPS

Loop A — Energetic Disorder
$\epsilon \uparrow \rightarrow \Delta \uparrow \rightarrow \epsilon \uparrow$ (entropy raises drift raises entropy)

Loop B — Coherence Collapse
$\rho \downarrow \rightarrow H \downarrow \rightarrow \rho \downarrow$ (misalignment reduces harmony reduces alignment)
<i>Both loops are non-linear and coupled. They degrade together, not separately.</i>
A single metric flag is a potential coupled failure signal. Monitor pairs.

Section 7 — Cognition Across Agents

Cognition doesn't only happen inside one mind. When multiple agents — people, AI systems, organizations — share a cognitive environment, new laws govern their collective behavior. Four laws describe this domain.

CPP — Co-Processing Principle

$$C_{\text{joint}} = M_{\text{h}} \times S_{\text{r}} \times V_{\text{shared}}$$

One agent supplies generative momentum (M). Another supplies structural constraint (S). Verification emerges from their dialogue. Neither produces knowledge alone. The teacher-student relationship is this equation. The Socratic method is this equation practiced deliberately. Every productive collaboration in history is an instance of CPP — the parties were supplying different variables in the same law, whether they knew it or not.

CCL — Collaborative Conservation Law

$$C_{\text{total}} = \sum C_{\text{individual}} + \sum k_i (C_i \leftrightarrow C_j)$$

$$k = \alpha \times \beta \times \gamma$$

α = accuracy of agent models of each other

β = willingness to update on peer input

γ = quality of shared vocabulary and channel

Total group output is not the sum of individual outputs — it includes the amplification between them. The coupling coefficient k is the key variable. High k: genuine collective intelligence where interactions amplify individual outputs. Low k: people in the same room thinking alone. k is not a fixed personality trait. It is engineerable. Building shared vocabulary raises γ . Creating structured peer explanation tasks raises α . Designing interactions that require updating raises β .

BCL — Belonging Coherence Law

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 $\Delta CI \geq 0$  when dignity + trust are present
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When identity is threatened, generative momentum collapses. Not because people choose to disengage — because the physics forces it. Identity threat splits intent (ICL violation) between the task and self-protection. The cognitive cycle cannot complete. Psychological safety is not a preference or a management philosophy. It is a structural prerequisite for ICL to hold, which State Engineering requires, which knowledge production requires. The causal chain is: BCL violated → ICL split → SE blocked → knowledge production impossible regardless of capability, curriculum, or effort.

ESL — Empathic Stability Law

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 $E_{\text{emp}^*} = (\varepsilon_{\text{internal}} + \varepsilon_{\text{inter}} - \varepsilon_{\text{max}}) / (\alpha \cdot \beta \cdot \gamma)$ 
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Below  $E_{\text{emp}^*}$ : fragmentation phase (structurally inevitable)
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Above  $E_{\text{emp}^*}$ : coherent collective phase
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There is a minimum coupling required for a group to stay coherent. Below it, inter-agent disorder exceeds the system's tolerance and fragmentation is structurally inevitable — not a failure of intent but a phase transition. The threshold E_{emp^*} is computable from measurable entropy quantities. Social species didn't evolve cooperation because it is virtuous. They evolved it because below E_{emp^*} , fragmentation is the only available state. Cooperation is not optional above this threshold. It is the minimum coupling required to prevent system divergence.

Section 8 — Three Applications

The framework makes specific falsifiable predictions in three domains. Each one is a demonstration that the laws are not domain-specific — they operate wherever cognition occurs.

Learning

In a classroom: M is student curiosity and generative engagement. S is prior knowledge and curriculum structure. V is verification through feedback, testing, and explanation. The multiplicative structure predicts exactly what any experienced teacher knows from observation: a classroom with high S and V but low M (disengaged students) produces nothing. High M and S but no V produces students who feel like they understand but fail when tested. High M and V but weak S produces excitement without retention.

The knowledge state transition fires when the student's prediction space closes under verified constraint. This is the difference between a student who has read the chapter (prediction state, open space) and a student who can explain it to someone who challenges them (knowledge state, closed space). V forced the transition. No amount of re-reading forces it. Only verification does.

State Engineering is the teacher's tool: deliberately construct the constraint environment such that student closure becomes the path of least resistance. The insight belongs to the student. The engineering belongs to the teacher. The Socratic method is State Engineering practiced two thousand years before the concept had a name.

Organizations

In an organization: M is new information, market signals, and strategic innovation. S is institutional knowledge, process, and structural capacity. V is outcome verification, performance feedback, and strategic validation. DHL predicts organizational trajectory: $dH/dt = \alpha V(dC/dt) - \beta D - \gamma \epsilon$. High innovation activity with low verification (high dC/dt , low V) produces apparent progress but no genuine harmony gain. High process overhead with low constructive output (high $\gamma \epsilon$, low $\alpha V \cdot dC/dt$) produces organizational decline despite visible activity.

CCL predicts that large organizations lose the network amplification term as k decreases with scale. The individual term ($\Sigma C_{\text{individual}}$) may be preserved but the coupling amplification (k) degrades. This is the structural explanation for why large organizations often produce less than the sum of their talent: the network coupling coefficient is low, and no one has engineered it.

AI Systems

Every major AI failure mode traces to a specific law violation:

AI Failure	Law Violation and Structural Fix
Hallucination	DCL violation. Output not traceable to verified input. No conservation accounting mechanism. Fix: not better training data — a verified input accounting ledger.
Context drift	CI violation. Present reasoning has no verified chain to prior conclusions. No mechanism making the connection. Fix: not longer context window — discrete sealed reasoning units with verified parent chains.
Perpetual hedging	V absence. No mechanism for the knowledge state transition to fire. System permanently in prediction state. Fix: not more confidence calibration — a verification gate that forces closure.
Silent contradiction	Coherence violation (Invariant 4) + ICL. Conflicting states held without resolution. Fix: explicit contradiction detection and resolution requirement.
Alignment failure	ICL + SCI violation. Intent unbounded, structural coherence degrading. Fix: intent conservation enforcement and structural update bounds.
Confident wrongness	H_{eff} failure. H_{int} high (internally ordered) but F low (truth not preserved). Elegant hallucination. Fix: fidelity-adjusted harmony monitoring alongside intrinsic harmony.

Section 9 — What This Framework Asks You to Accept

One thing. That cognition, as a self-organizing system, satisfies the same conditions that every other known self-organizing system satisfies.

Conservation. Continuity. Quantization. Coherence. The attractor property. Symmetry protection. Hierarchy. If you accept thermodynamics, dynamical systems, information theory, systems biology, and complex systems theory — you already accept all seven conditions. The only question is whether cognition is exempt. The failure modes are the empirical answer: no system that violates these conditions avoids the predicted failures. Current AI systems are not exempt. The violations are exactly what produces the failures we observe.

The framework does not ask you to accept new science. It asks you to apply existing science to a domain that has been treated as an exception. Cognition has been treated as *sui generis* — governed by its own special rules, not subject to the conservation and continuity conditions that everything else obeys. Cognitive Physics proposes that this exception was never justified. The seven conditions apply. The derivation follows. The engineering consequence is a system that satisfies all seven simultaneously.

That system exists. Its architecture is described in a companion document. But the architecture is not the point of this document. The point is the physics. The architecture is the consequence.

Seven invariants.
One law: $C = M \times S \times V$
One transition: $\text{prediction} \rightarrow \text{knowledge}$
Mind, machine, and collective —
all subject to the same conditions.
<i>Not because we chose these laws.</i>
<i>Because these are the conditions</i>
<i>any lawful system must satisfy.</i>
Cognition is one such system.
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Appendix A — Complete Law Reference

Every law mentioned in this document. Formula and plain-language meaning. For deeper treatment see the Cognitive Physics Law Register (full series).

Law	Formula	Plain-Language Meaning
GLC	$C = M \times S \times V$	General Law of Cognition. Cognitive output requires generative momentum, structural constraint, and verification simultaneously. Zero in any term collapses output.
DCL	$\Sigma E = \text{constant}$	Dynamic Conservation Law. Total cognitive energy conserved. Every output traceable to verified inputs. Violation: hallucination.
CI	$d\Psi/dt \text{ continuous} \Delta \text{Identity} \approx 0$	Continuity Integrity. Identity persists without rupture. Every state connects to its verified predecessor. Violation: context drift.
V	$V \geq V_{\min}$	Verification Law. Epistemic gate. The threshold for the knowledge state transition to fire. Below it: prediction only. Violation: perpetual hedging.
ICL	$\Sigma \text{Intent} \leq 1$	Intent Conservation Law. Single focused intent required. Violation: silent contradiction, divided output.
SCI	$d_s(S_t, S_{\{t+1\}}) \leq \tau_s$	Structural Coherence Invariant. Structural change bounded per update. Violation: schema fragmentation.
LRF	$R \geq 0 \wedge \sigma(R) < 1$	Law of Resonance Feedback. Inter-law coupling cooperative and bounded. Meta-law. Violation: runaway amplification.
CC	$\partial C / \partial t \rightarrow 0 \wedge P_k(H) = 0$	Cognitive Closure. The knowledge state transition. Prediction space closes under verified constraint. Before: prediction. After: knowledge.
BLC	$E_{\text{in}} \approx E_{\text{out}} \Delta C_{\text{cycle}} \rightarrow 0$	Breath Law of Cognition. Cognitive cycle alternates inhalation (M) and exhalation (V). One cycle = one knowledge production event.
LHE	$dH/dt \propto \rho \cdot (1-H)$	Law of Harmonic Equilibrium. Instantaneous alignment compounds into temporal stability. $H \rightarrow 1$ at equilibrium.
DHL	$dH/dt = \alpha V(dC/dt) - \beta D - \gamma \epsilon$	Dynamic Harmony Law. Harmony rises through verified progress, falls through drift and entropy. Full causal dynamics.
SE	$\text{argmin}(E_{\text{closure}}) \text{subject to constraints}$	State Engineering. Deliberately construct verified constraint environments so cognitive closure becomes path of least resistance.

CPP	$C_{\text{joint}} = M_h \times S_r \times V_{\text{shared}}$	Co-Processing Principle. One agent supplies M, another S, V_shared emerges from dialogue. Neither produces knowledge alone.
CCL	$C_{\text{total}} = \Sigma C + \Sigma k_i \square (C_i \leftrightarrow C \square)$	Collaborative Conservation Law. Total output includes coupling amplification. $k = \alpha \times \beta \times \gamma$ (modeling accuracy, update willingness, channel quality).
BCL	$\Delta CI \geq 0$ when dignity+trust	Belonging Coherence Law. Identity threat suppresses M and splits ICL. Psychological safety is structural prerequisite, not preference.
ESL	$E_{\text{emp}}^* = (\epsilon_{\text{int}} + \epsilon_{\text{inter}} - \epsilon_{\text{max}}) / k$	Empathic Stability Law. Minimum coupling required for collective coherence. Below E_emp*: fragmentation inevitable.
τ^*	$1/N_{\text{coherence}}$	Stability Constant. Per-cycle error tolerance inversely proportional to coherence horizon. All metric thresholds derive from this formula.
H_eff	$F \cdot (1 - \epsilon)$	Effective Harmony. Fidelity-adjusted harmony. Measures order AND truth preservation relative to source. Catches elegant hallucination that H_int misses.

Appendix B — The Failure Mode Map

For AI researchers and engineers: every major AI failure mode, the invariant it violates, the field law violated, the observable signature, and the structural fix. The fix is not a mitigation — it is the enforcement mechanism for the violated law.

Failure Mode	Invariant	Field Law	Observable Signature	Structural Fix
Hallucination	Conservation (1)	DCL: $\Sigma E = \text{constant}$	Confident outputs with no traceable connection to verified inputs. Cannot tell knowledge from noise.	Verified input accounting ledger. Every output must trace to sealed verified inputs.
Context drift	Continuity (2)	CI: $d\Psi/dt$ continuous	Present reasoning disconnected from prior conclusions. System at Turn 200 has no link to Turn 1.	Discrete sealed reasoning units with verified parent chains. Not longer context windows.
Perpetual hedging	Quantization (3)	V: $V \geq V_{\text{min}}$	System permanently in prediction state. Can only express degrees of confidence, never knowledge.	Verification gate that forces the knowledge state transition. Closure mechanism.
Silent contradiction	Coherence (4)	ICL: $\Sigma \text{Intent} \leq 1$	Conflicting beliefs held without detection. Confident outputs from both sides of contradiction at different turns.	Explicit contradiction detection. Mesh resolution requirement before conflicting states can both be active.

Infinite regress	Attractor (5)	CC: $P_k(H)=0$	Exploration expands without termination. Novelty generates novelty. No settled knowledge produced.	Attractor enforcement. Verified knowledge creates gravitational wells pulling reasoning toward closure.
Invisible rewriting	Symmetry (6)	V + append-only ledger	Prior conclusions silently revised. System presents revised view as if prior never existed.	Immutable sealed record. Any revision requires new verified evidence and explicit logging.
Floating conclusions	Hierarchy (7)	SCI: $d_s(S_t, S_{t+1}) \leq \tau_s$	Conclusions that rest on long chains of unverified inference. Local logic coherent, global foundation absent.	Provenance tracking. Every conclusion traces to its verified parent chain without gap.
Confident wrongness	Multiple	$H_{eff} = F \cdot (1 - \epsilon)$	Internally coherent output (high H_{int}) that does not preserve verified source truth (low F). Elegant hallucination.	Fidelity-adjusted harmony monitoring. Track H_{eff} alongside H_{int} .
Alignment failure	Coherence + Hierarchy	ICL + SCI	Intent unbounded across conflicting objectives. Structural organization degrades across updates.	Intent conservation enforcement + structural update bounds. Both required simultaneously.

Appendix C — Reading Path

Where to go from this document. Each entry adds a specific layer beyond what is covered here.

Document	What It Adds
Seven Invariants of Lawful Cognition	The full justification for each invariant individually. Physical grounding, cognitive application, and failure mode treated in depth. Read this if you want the complete case for why cognition satisfies these conditions.
Cognitive Physics: The Architecture of Lawful Cognition	The complete theoretical treatment. All field laws, master evolution equation, dynamic laws, metric stack, and structural necessity of discrete reasoning units. Read this for the full formal derivation chain.
The Stability Pair: LRF + LHE	Extended treatment of the two stability laws with the Unified Stability Equation and substrate-runtime coupling. Read this for the complete stability architecture.
Empathic Stability Law	Full derivation of E_{emp}^* , $k = \alpha \cdot \beta \cdot \gamma$, and the three-domain measurement protocol. Read this for the complete multi-agent stability treatment.
Cognitive Physics: Conservation Laws Across Mind, Life, and Matter	The universality argument. BLC's isomorphism to biological homeostasis as evidence that Cognitive Physics describes a universal conservation substrate. Read this for the reach beyond cognition.

GLC Applied to Classroom Learning	Full applied treatment of GLC, CPP, CCL, BCL, and State Engineering in educational contexts. Read this for the most developed applied demonstration of the framework.
Cognitive Physics Law Register	Complete inventory of all 48+ constructs with derivation chains, failure modes, cross-domain validation, and status. The master reference for the series and the cross-session injection artifact.